

Physics 5A Discussion (Week 0)

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Example: We know 1 foot is 0.305 meters. Denote “ft” for feet and “m” for meters, so

$$100 \text{ ft}^2 = (10 \text{ feet}) \cdot (10 \text{ feet}) = (3.05 \text{ meters}) \cdot (3.05 \text{ meters}) = 9.30 \text{ m}^2.$$

Problem 1: The radius of a disc is 5 meters. The area of a disc is $A = \pi r^2$ where r is the radius. What is its area in square feet (ft^2)?

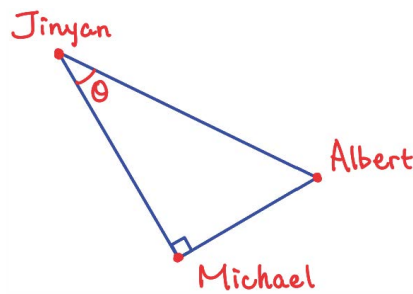
Example: If the force F acting on an object of mass m (measured in kilograms) and acceleration a (measured in meters per second squared) obeys the formula $F = ma$, then the unit of the force is

$$[F] = [m] \cdot [a] = (\text{kilogram}) \cdot (\text{meters per second squared}) = \text{kg m} / \text{s}^2.$$

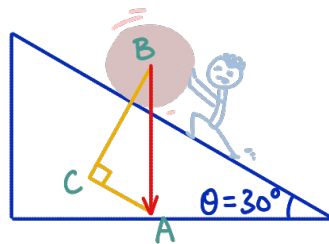
Later on we will call this new unit (kg m/s^2) as Newton (N). The letter m is italicized when we use it to denote mass, but m is not italicized when we use it to denote the unit meter.

Problem 2: Knowing that force F is measured in kg m/s^2 , density ρ is measured in kg/m^3 , and velocity v is measured in m/s , find the unit of the quantity B in the formula

$$F = \rho v^2 B.$$



Problem 3: (Make sure you include units in your final answers!) Using the figure shown, if the distance between Jinyan and Michael is 4 meters, and the distance between Michael and Albert is 3 meters, what is the distance between Jinyan and Albert? What is the angle θ in degrees? What is the angle θ in radians?



Problem 4: Jinyan is pushing a huge rock up an incline, which makes an angle 30° above the horizontal. His friend Michael is analyzing the force Jinyan must apply to hold the rock, using the geometry shown in the figure. There is a small right triangle $\triangle ABC$. The line segment AC is parallel to the incline, and the line segment BC is perpendicular to the incline. If the force arrow BA has a “length” of 100 Newtons, find the “lengths” of the line segments AC and BC . All these “lengths” are measured in Newtons, so your final answer should be expressed with the unit N. (*Hint:* First find $\angle ABC$ using the given information, then use the sine and cosine functions to find the lengths of the sides of the right triangle.)

Problem 1

I recommend first converting everything to the desired unit and then doing the algebra and computations, so we first write

$$r = 5 \text{ m} = \frac{5}{0.305} \text{ ft},$$

then we calculate the area

$$A = \pi r^2 = \pi \cdot \left(\frac{5}{0.305} \text{ ft} \right)^2 = \pi \cdot \left(\frac{5}{0.305} \right)^2 \text{ ft}^2 \approx 844 \text{ ft}^2.$$

Make sure you have the correct unit in your final answer. For this class, when it comes to decimal places, I recommend keeping 3 or 4 significant digits unless specified otherwise.

Problem 2

We want to find the unit of B , and the units of all other quantities are known, so we rearrange the $F = \rho v^2 B$ formula to isolate B ,

$$B = \frac{F}{\rho v^2},$$

then we just plug in the units and get

$$[B] = \frac{\text{kg m/s}^2}{(\text{kg/m}^3) \cdot (\text{m/s})^2} = \text{m}^2.$$

Now we see that B has the unit of m^2 , which is something like an “Area”.

Problem 3

The angle at Michael is a right angle, so the distances from Jinyan to Michael and from Michael to Albert are the two sides of a right triangle. The distance from Jinyan to Albert is the hypotenuse. Using the Pythagorean theorem, we have

$$(\text{Jinyan-Michael})^2 + (\text{Michael-Albert})^2 = (\text{Jinyan-Albert})^2.$$

Then we plug in the two known distances and get

$$\begin{aligned} (4 \text{ m})^2 + (3 \text{ m})^2 &= (\text{Jinyan-Albert})^2, \\ \text{Jinyan-Albert} &= \sqrt{(4 \text{ m})^2 + (3 \text{ m})^2} \\ &= \sqrt{16 + 9} \text{ m} = 5 \text{ m}. \end{aligned}$$

To find the angle θ at Jinyan, the 3 m side is opposite to θ and the 4 m side is adjacent to θ , so we use the tangent function,

$$\tan(\theta) = \frac{\text{opposite}}{\text{adjacent}} = \frac{3 \text{ m}}{4 \text{ m}} = \frac{3}{4}.$$

Taking the inverse tangent of both sides gives the angle in degrees,

$$\theta = \arctan\left(\frac{3}{4}\right) = 36.9^\circ.$$

Finally, we convert this angle from degrees to radians,

$$\theta = 36.9^\circ \cdot \frac{\pi \text{ rad}}{180^\circ} = 0.644 \text{ rad}.$$

Thus the distance between Jinyan and Albert is 5 m, and the angle is 36.9° , or equivalently 0.644 rad.

Problem 4

The line segment BA is vertical, so it makes an angle 90° with the horizontal. The line segment AC is parallel to the incline and makes an angle 30° with the horizontal. Therefore, the angle at A in the small right triangle is

$$\angle BAC = 90^\circ - 30^\circ = 60^\circ.$$

Since $\angle ACB = 90^\circ$, the three angles of the triangle give

$$\angle ABC = 180^\circ - \angle BAC - \angle ACB = 180^\circ - 60^\circ - 90^\circ = 30^\circ.$$

The line segment BA is the hypotenuse with a “length” of 100 N. Relative to $\angle ABC = 30^\circ$, the line segment AC is the opposite side, so

$$\sin(30^\circ) = \frac{AC}{BA} = \frac{AC}{100 \text{ N}}.$$

We thus find

$$AC = (100 \text{ N}) \sin(30^\circ) = (100 \text{ N}) \left(\frac{1}{2}\right) = 50.0 \text{ N}.$$

The line segment BC is the adjacent side, so

$$\cos(30^\circ) = \frac{BC}{BA} = \frac{BC}{100 \text{ N}}.$$

Therefore,

$$BC = (100 \text{ N}) \cos(30^\circ) = (100 \text{ N}) \left(\frac{\sqrt{3}}{2}\right) = 86.6 \text{ N}.$$

Thus the force component parallel to the incline has a “length” $AC = 50.0 \text{ N}$, while the force component perpendicular to the incline has a “length” $BC = 86.6 \text{ N}$.